

# Lecture 13 – Space complexity. Circuit complexity.

NTIN071 Automata and Grammars

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*\* Adapted from the Czech-lecture slides by Marta Vomlelová with gratitude.  
The translation, some modifications, and all errors are mine.*

## Recap of Lecture 12

- time complexity, for TM as well as NTM
- the class P
- the class NP: verifier-based and nondeterminism-based definitions
- polynomial-time reductions
- NP-complete problems
- Cook-Levin theorem: SAT is NP-complete
- the class co-NP
- exponential time complexity

# Space complexity

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# Space complexity

Similarly to time, measure space required for computation:

## Definition

The **space complexity** of a [deterministic, single-tape] Turing machine that always halts is  $f : \mathbb{N} \rightarrow \mathbb{N}$ , where  $f(n)$  is the maximum number of cells that  $M$  accesses on any input of size  $n$ .

For nondeterministic Turing machines we take the maximum over all computation paths.

But this does not work for **sublinear** space complexity, in particular, **logspace** (**L**) and **nondeterministic logspace** (**NL**).

The solution is to have two tapes: a read-only input tape, and a working tape whose space we measure. (If we want output, then a third 'write-only' output tape, only traverse in one direction.)

# Classes of space complexity

For  $f : \mathbb{N} \rightarrow \mathbb{R}^+$ , define the space complexity classes:

$\text{SPACE}(f(n)) = \{L \mid L \text{ decidable by a DTM in space } O(f(n))\}$

$\text{NSPACE}(f(n)) = \{L \mid L \text{ decidable by an NTM in space } O(f(n))\}$

$\text{L} = \text{SPACE}(\log(n))$       note:  $\log(n^k) \in O(\log(n))$

$\text{NL} = \text{NSPACE}(\log(n))$

$\text{PSPACE} = \bigcup_k \text{SPACE}(n^k)$

$\text{NPSPACE} = \bigcup_k \text{NSPACE}(n^k)$

## Relationship between space and time

- Trivially,  $\text{TIME}(f(n)) \subseteq \text{SPACE}(f(n))$  and  $\text{NTIME}(f(n)) \subseteq \text{NSPACE}(f(n))$  (at most one cell per step)
- $\text{NTIME}(f(n)) \subseteq \text{SPACE}(f(n))$  (DFS on computation graph, store path from root as sequence of choices, not configs)
- If  $f(n) \geq \log(n)$ , then  $\text{NSPACE}(f(n)) \subseteq \text{TIME}(2^{O(f(n))})$  (only  $2^{O(f(n))}$  configs, fully construct computation graph + run BFS)

$$L \subseteq NL \subseteq P \subseteq NP \subseteq PSPACE \subseteq NPSPACE \subseteq EXPTIME$$

Which inclusions are equalities? Which inclusions are strict?

- $PSPACE = NPSPACE$  (Savitch's theorem, coming up)
- $P \neq EXPTIME$  (Time hierarchy theorem, omitted)
- the rest is wide open

## Example: NFA accepts everything?

$$\text{ALL}_{\text{NFA}} = \{ \langle A \rangle \mid A \text{ is an NFA such that } L(A) = \Sigma^* \}$$

We can decide  $\overline{\text{ALL}_{\text{NFA}}}$  in **nondeterministic linear space**:

- markers on states of  $A$  to keep track of set of possible states
- start with marker on start state
- repeat  $2^{|Q|}$  times:
  - guess an input symbol  $a \in \Sigma$
  - move markers according to transitions (simulate reading  $a$ )
  - if no marker on accepting state, accept
- if not accepted so far, reject
- note:  $2^{|Q|}$  is enough, more means a loop that can be shortened

Thus  $\text{ALL}_{\text{NFA}} \in \text{co-NSPACE}(n) \subseteq \text{co-SPACE}(n^2) = \text{SPACE}(n^2)$

Is  $\text{ALL}_{\text{NFA}}$  in  $\text{NP} \cup \text{co-NP}$ ?

## Savitch's theorem

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# Savitch's theorem

## Theorem (Savitch)

For any  $f : \mathbb{N} \rightarrow \mathbb{R}^+$  with  $f(n) \geq \log(n)$ :

$$\text{NSPACE}(f(n)) \subseteq \text{SPACE}(f(n)^2)$$

## Corollary

$$\text{PSPACE} = \text{NPSPACE}$$

**Idea:** Simulate the NTM deterministically by checking reachability in the configuration graph using a recursive algorithm that uses only  $O(f(n)^2)$  space.